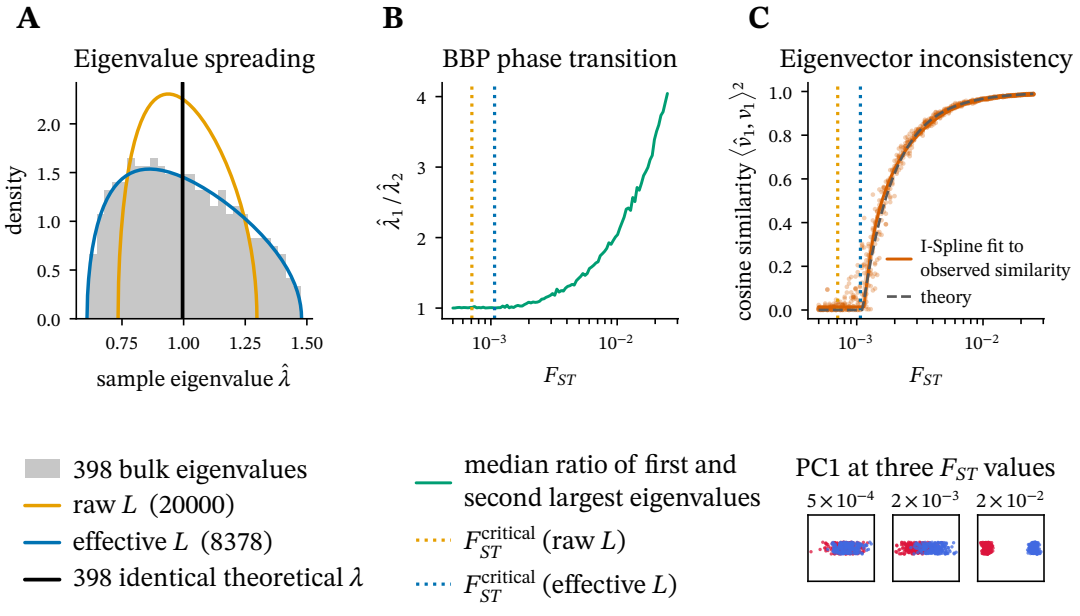

A
Eigenvalue spreading
density
sample eigenvalue $\hat{\lambda}$
B
BBP phase transition
$\hat{\lambda}_1 / \hat{\lambda}_2$
F_{ST}
C
Eigenvector inconsistency
cosine similarity $\langle \hat{v}_1, v_1 \rangle^2$
I-Spline fit to observed similarity
theory
F_{ST}
398 bulk eigenvalues
raw L (20000)
effective L (8378)
398 identical theoretical λ
median ratio of first and second largest eigenvalues
F_{ST}^{critical} (raw L)
F_{ST}^{critical} (effective L)
PC1 at three F_{ST} values
5×10^{-4}
2×10^{-3}
2×10^{-2}